Adithya Sathyanarayana

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Research Interests

Arithmetic geometry: Galois module structure and Deligne–Lusztig varieties; computational aspects of Drinfeld modules; arithmetic statistics of class groups of number fields.

Education

Michigan State University, United States — Expected June 2027
Ph.D. in Mathematics
Advisor: Georgios Pappas
Dissertation: *Galois module structure of Deligne–Lusztig varieties* (in progress)

IISER Tirupati, India — June 2022
Indian Institute of Science Education and Research Tirupati
B.S.–M.S. Dual Degree
Advisor: D. S. Nagaraj
Master's thesis: *Class field theory*

Research in Preparation

Galois module structure of Deligne–Lusztig varieties for $\mathbf{G} = \mathrm{GL}_3$
In preparation. Research webpage

Research Projects

Arithmetic statistics of p-torsion in class groups of quartic number fields — 2026–present
Participant in Sameera Vemulapalli's project group at Arithmetic Geometry at UNT 2, University of North Texas (August 3–7, 2026).
Studying p-torsion in class groups of quartic number fields admitting an integral basis of the form $1, x, y, xy$, in connection with Bhargava's parametrizations of quartic rings. Some Magma code and preliminary analysis can be found in the following project repository:
https://github.com/s-adithya/agunt2_project
Ongoing project.

Efficient computation of the Weil pairing for Drinfeld modules — 2026–present
Participant in Antoine Leudière's project group at Rethinking Number Theory, held online June 22–July 3, 2026.
Working toward explicit computation of Weil pairings for Drinfeld modules through implementations of pairing algorithms in SageMath or other open-source software. Ongoing project.

Invited Talks

Galois module structure of Deligne–Lusztig varieties for $\mathbf{G} = \mathrm{GL}_3$
Group, Lie and Number Theory Seminar
University of Michigan (Invited by Dr. Charlotte Chan) November 2026 (scheduled)

Galois module structure of Deligne–Lusztig varieties for $\mathbf{G} = \mathrm{GL}_3$
Graduate Student Lecture Series (UBDoG)
University at Buffalo October 2026 (scheduled)

Galois module structure of Deligne–Lusztig varieties for $\mathbf{G} = \mathrm{GL}_3$*: twisted and untwisted cases*
AMS Special Session on Springer Fibers, Harmonic Analysis, and Arithmetic
George Washington University October 4, 2026

Galois module structure of Deligne–Lusztig varieties for $\mathbf{G} = \mathrm{GL}_3$
Automorphic Forms and Representation Theory Seminar
Purdue University (Invited by Dr. Daniel Le) September 2026

Galois module structure of Deligne–Lusztig varieties for $\mathbf{G} = \mathrm{GL}_3$
Graduate Student Conference in Algebra, Geometry, and Topology (GTA Philly)
Temple University May 30, 2026

Contributed Talks

Galois module theory
Student Algebra Seminar
Michigan State University Fall 2026

Deligne–Lusztig varieties
Student Algebra Seminar
Michigan State University October 2025

Non-split Cartan subgroups and the associated modular curves
Learning Seminar: The Class Number One Problem d'après Serre
Michigan State University Spring 2025

Reciprocity laws in number theory
Student Algebra Seminar
Michigan State University 2024

Poster Presentations

Galois module structure of Deligne–Lusztig varieties for $\mathbf{G} = \mathrm{GL}_3/\mathbb{F}_p$*: untwisted and twisted cases*
The Australian Direction (invited poster)
University of Utah August 11, 2026

Galois module structure of Deligne–Lusztig varieties for $\mathbf{G} = \mathrm{GL}_3$
Algebraic Geometry Northeastern Series (AGNES)
Brown University November 21, 2026 (scheduled)

Teaching Experience

Instructor of Record, Michigan State University

MTH 103A: College Algebra I	Spring 2025
MTH 234: Multivariable Calculus	Fall 2024
MTH 314: Matrix Algebra with Applications	Summer 2024
MTH 124: Survey of Calculus (online)	Summer 2023

Teaching Assistant, Michigan State University

MTH 317H: Honors Linear Algebra	Fall 2026
MTH 299: Introduction to Proofs	Spring 2026
MTH 818: Graduate Algebra I (qualifying examination course)	Fall 2025
MTH 314: Matrix Algebra with Applications	Spring 2024
MTH 234: Multivariable Calculus	Fall 2023

Teaching Assistant, IISER Tirupati

Introduction to Proofs	Monsoon 2021

Teaching Leadership and Support

Michigan State University

Lead Teaching Assistant Fall 2024, Spring 2025
Conducted classroom observations and provided teaching feedback to 8 teaching assistants and 16 undergraduate learning assistants.

Panelist at orientation for incoming Ph.D. students Fall 2024
Helped incoming students transition into teaching and academic life at MSU.

Tutor and Supervisor, Mathematics Learning Center 6+ semesters
Assisted students with mathematics coursework and supervised peer tutors.

Undergraduate Mentoring

MSU Summer Topology Program Summer 2026
Mentored two undergraduate students on the following projects:

- Jack Finlan-Hitt: Euler characteristics and the Lefschetz fixed-point theorem in algebraic topology
- Jacob Kantzer: The Gauss–Bonnet theorem and the classification of surfaces

Pedagogical and Professional Development

Certification in College Teaching Expected Spring 2027
Michigan State University

- Coursework: MTHE 879, Teaching College Mathematics
- Completed the Certification in College Teaching Institute Workshop (May 2026)
  - Selected workshop topics: institutional contexts and missions; effective learning environments; assessment of student learning; technology in teaching

Erdős Institute Data Science Boot Camp Completed Summer 2026
Project: *Who Bears the Congestion Price?*
Project repository

Seminar Organization

Student Algebra Seminar, Michigan State University Fall 2024, Spring 2025
Co-organizer. Seminar website

Seminar Participation

Michigan State University

Agentic AI Seminar	Fall 2026
Lean Seminar	Spring 2026, Fall 2026
Étale Cohomology Learning Seminar	Fall 2025, Spring 2026
Learning Seminar on the Class Number One Problem	Spring 2025
Class Field Theory Learning Seminar	Summer 2023

Service

Treasurer and Member 2024–present
Association for Women in Mathematics, MSU Student Chapter

Department of Mathematics Representative and Council Member 2025–2026
Council of Graduate Students (COGS), Michigan State University

International Student Liaison and Member 2022–2023
American Mathematical Society, MSU Student Chapter

Sports Secretary, IISER Tirupati 2020–2021

Member, Hostel and Dining Committee, IISER Tirupati 2018–2019

Awards

Douglas A. Spragg Endowed Fellowship Award — 2026
Department of Mathematics, Michigan State University

College of Natural Science Dissertation Completion Fellowship — Summer 2026
Michigan State University

Certificate for Outstanding Performance in Extracurricular (Cultural) Activities — 2018
Awarded by the Director of IISER Tirupati

Travel Grants

AMS Sectional Meeting Travel Grant — October 2026
To attend the 2026 Fall Eastern Sectional Meeting

COGS Fellowship, Michigan State University — March 2026
To attend the Arizona Winter School

Conferences and Workshops

Individual Conferences and Workshops

The Australian Direction — August 10–14, 2026
A conference honoring the mathematical contributions of Matt Emerton and Mark Kisin
University of Utah, Salt Lake City, UT

Arithmetic Geometry at UNT 2 (UNT2) — August 3–7, 2026
University of North Texas, Denton, TX

Rethinking Number Theory (RNT) — June 22–July 3, 2026
Online

Arizona Winter School 2026 — March 2026
Computational Aspects of Arithmetic Geometry and Cryptography
University of Arizona, Tucson, AZ
Participant in Sara Arpin's study group on *Exploiting higher dimensions in isogeny-based cryptography*

CMND 2025 Thematic Program — June 9–13, 2025
Discrete Groups in Topology and Algebraic Geometry
University of Notre Dame, Notre Dame, IN

Arizona Winter School 2024: Abelian Varieties — March 2024
University of Arizona, Tucson, AZ

Graduate Combinatorics, Algebra, and Topology Conference — November 2023
Binghamton University, Binghamton, NY

Preliminary Arizona Winter School 2023 — October 2023
Elliptic Curves and Abelian Varieties (online)
Mini-course: *Abelian Varieties over Finite Fields*, Lassina Dembele

Preliminary Arizona Winter School 2022 October 2022
Online
Mini-course: *Heights in Diophantine Geometry*, Padmavathi Srinivasan

Recurring Conferences and Regional Meetings

Michigan Number Theory Day
Michigan State University, East Lansing, MI: Fall 2026
University of Michigan, Ann Arbor, MI: Fall 2025

Michigan Algebraic Geometry Symposium (MAGS)
Michigan State University, East Lansing, MI: Spring 2026, Fall 2023
University of Michigan, Ann Arbor, MI: Fall 2025, Spring 2023

Midwest Arithmetic Geometry and Number Theory Series (MAGNTS)
Michigan State University, East Lansing, MI: Spring 2026, Fall 2023
University of Michigan, Ann Arbor, MI: Fall 2025, Spring 2023
The Ohio State University, Columbus, OH: 2025
University of Illinois Chicago, Chicago, IL: Fall 2022

Graduate Student Conference in Algebra, Geometry, and Topology (GTA Philly)
Temple University, Philadelphia, PA: Spring 2026, Fall 2025, Fall 2024

Technical Skills and Languages

- Programming: Python, MATLAB, JavaScript
- Mathematical software: Macaulay2, SageMath, Magma, LEAN
- Data tools: DuckDB, GeoPandas, scikit-learn, statsmodels, Plotly
- Other technical tools: LaTeX, HTML, CSS, Linux, GitHub, Conda
- Languages: Kannada (native), English (proficient), Hindi (familiar), French (beginner)